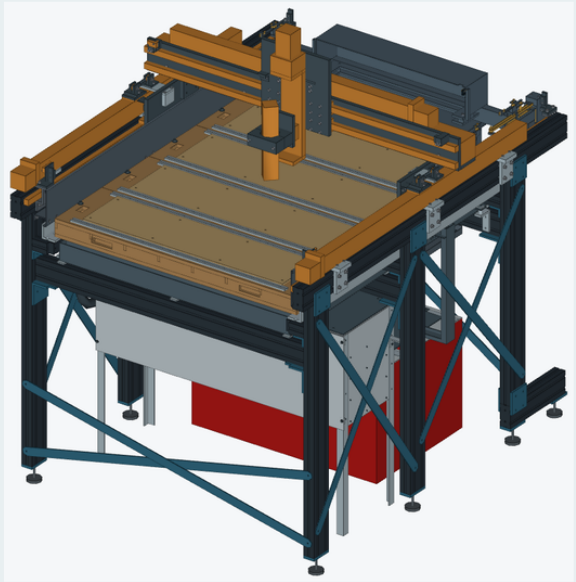
4080 table. Independent beds. Retractable ATC.

Current small machine: 800 X / 1,000 Y / 200 Z nominal travel. New Z hardware is required.

XR2 | 4080 WOOD + PLASMA TABLE

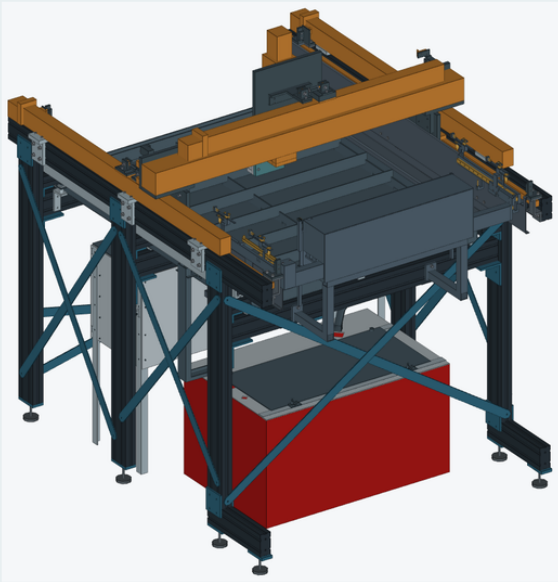
800 X / 1,000 Y / 200 Z nominal travel • Engineering revision

01 WOODWORKING



Removable dry bed with aluminum T-slot workholding.

02 PLASMA



Steel hot zone; dry bed in covered storage.

03 FRAME



Six continuous posts; no second full-length frame tier.

Geometry, loads and conversion paths are checked separately.

Read the current ATC fit result and physical release register before fabrication. No prices.

Actual CAD solids shown. Tool, torch and cable envelopes remain subject to measurement.

Actual generated CAD. Owned electronics enclosure stays excluded; the red box is the water receiver.

Structure, workholding and water

Pan loads and router loads have separate bearing paths. Component calculations remain conditional.

Frame

12.3574 m finished 4080; 19 cuts

Supports

Six posts; rear bases; eight main feet

Dry carriers

Four 932 x 257 x 70.8

Wear field

770 x 960 enclosing outline; keep-outs apply

Box construction

12.7 skins / 39.05 core / 6.35 wear

Hot pan

894 x 1096 outside; 1.9 nominal plain steel

Slats

Eight 880 x 40 x 1.9; center-keyed / free ends

Owned aluminum

Two 304.8-square blanks; 83 nested blanks

WATER VOLUMES

Operating level	**37.05 L**
Overflow starts	**45.68 L**
Brim	**56.23 L**
Gravity-drain residue	**0.45 L**

Require 65 L measured usable receiver capacity. The nominal 20-gallon storage label does not verify its liquid load, useful volume or temperature suitability.

100 N VERTICAL SENSITIVITY

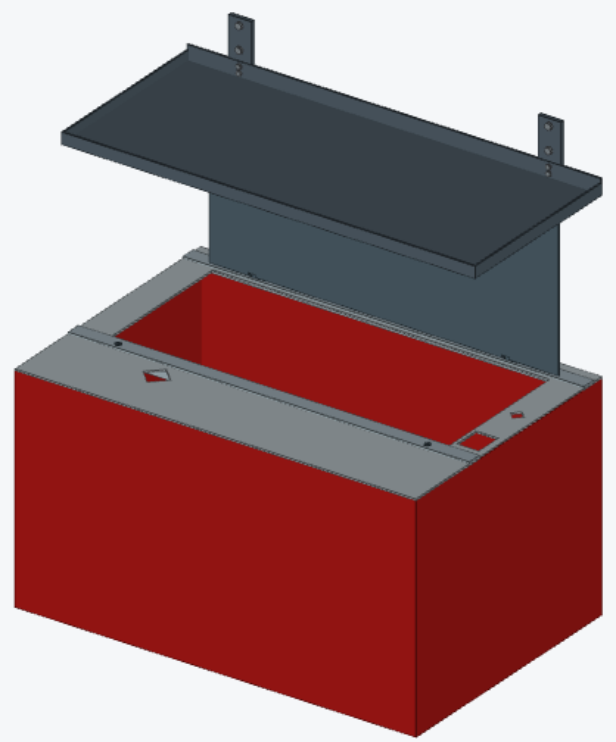
Selected material case: 0.233 mm bed + ledger + upper-beam subtotal against a 0.30 mm framework/bed target. Weaker plywood case: 0.314 mm, exceeding the target before omitted joints.

Half-lap integration is a nominal composite-beam sensitivity. Local notch/glue transfer, joints and actual tool-loop compliance are unqualified; this is not assembled-frame FEA.

Section properties, plywood direction, adhesive quality and actual joint preload are conditions of the estimates.

Service access and manufacturable parts

Routine cleaning stays in place. Rare receiver replacement has a separate reversible disassembly sequence.



OPEN / LOCKED CLEAN-OUT HATCH

Conservative reservoir envelope and modeled service hardware. Surrounding machine remains in the collision check. Metal cross-angles and rim feet support the hatch independently of the cut plastic lid.

Open assembly incl. locks Z689.3; roof Z700: 10.7 mm allowance before support/thermal effects.

CONVERSION AND MAINTENANCE

32 complete cassette-transfer segment enclosures are checked inside the plan.

Remove the complete installed tool module, cassette clamps and plasma-only ledger armors; retain Z and park the ATC before conversion.

A 450 mm cleaning brush reaches the receiver floor through the framed hatch. Two index pins hold the cover open.

Rare tank replacement removes the bolt-on ATC module and specified guards/ledgers; HGR motion rails stay installed. Temporary supports and two-person handling require physical qualification.

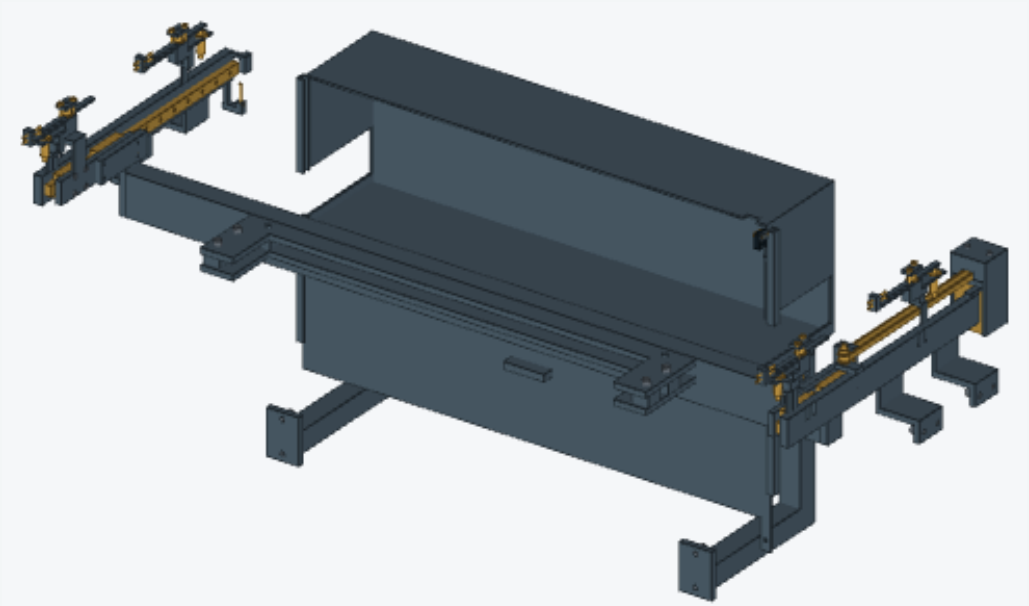
FABRICATION FILES

174 custom frame/motion STEP files with read-back checks; 42 labeled plate cross-section DXFs; 65 bed/steel profile DXFs. Shield and ATC schedules add their own files.

Reference plate sections do not replace counterbore, countersink, tap, edge-drilling or bend instructions. No production G-code is supplied.

Full-bed ATC and the 200 mm Z redesign

Nominal spindle reference, not a measured nut plane. Magazine dimensions are an acceptance specification.



ATC MECHANISM / DEPLOYED

200 mm carrier stroke; 335 mm guides.

Magazine base Z1030.15; stock + clamps <= 40.

Retained projection below base <= 26.35.

50 mm tool projection: flyover 9.85; minimum 5.

Worst pocket beam: 0.087 mm, excluding joints.

DIGITAL REVIEW: PASS WITHIN REPORT SCOPES

Match the selected ZXHT80-200 mounting/output, shaft/coupler and power-off retention interfaces. The brake-motor provision is conditional on actual fit and dynamic testing.

Verify the real magazine envelope, pocket/nut plane, bolt pattern, retained tools, spindle reverse/control and loaded docking repeatability. No successful M6 cycle is asserted.

Confirm the actual extrusion, nuts, fasteners, guide alignment and screw lead. Measure assembled framework/bed and complete spindle-to-work compliance separately.

Cold-test drain/overflow and receiver capacity, then qualify guard/storage temperatures. Trial conversion and rare service with final hoses, tooling and measured masses.

The measurement worksheet separates these real interfaces and tests from modeled geometry. This revision does not qualify steel milling.

USD 1,000 remains the complete-build constraint. Replacement Z and ATC hardware are not assumed owned.

No new pricing or cart changes. Sources: current CAD/check reports and preserved primary supplier drawings.

RapidChange setup FAQ